

Title

Monoclonal antibody L9LS demonstrates partial protective efficacy against malaria parasitaemia in young East African children

Abstract

Background: Monoclonal antibody-based malaria prophylaxis represents an emerging intervention strategy. We evaluated safety and efficacy of L9LS in an area of intense year-round transmission. **Methods:** Following safety/tolerability assessment in age-stratified cohorts, 324 children aged 5-59 months in western Kenya were randomized to receive two L9LS injections (10-20 mg/kg at months 0 and 6; n=106), single injection (n=108), or placebo (n=110). Children underwent monthly blood-slide examination for 12 months. **Results:** In the two-dose group, 66% of children experienced at least one episode of parasitaemia during follow-up versus 83% in placebo recipients (protective efficacy 42.7%; 95% CI 22.5-57.7; p=0.0003). Adverse events were rare and balanced between groups (grade 3 or worse in 1% of L9LS injections versus 1% of placebo). No serious safety signals emerged. Peak antibody levels correlated with parasite suppression. **Conclusions:** Bi-annual L9LS dosing conferred modest but clinically relevant malaria protection in young children, with an acceptable safety profile. Higher dosing strategies warrant exploration to improve efficacy in this high-transmission setting.